Fractions, Decimals and Percentages — Mark schemes

PAPERS A TO D · 30 MARKS EACH · TUTOR COPY

Marks in square brackets show how the total is split. Where a question is worth more than one mark, award method marks even when the final answer is wrong — and take them away for a correct answer with no working only if the question said "show your working". The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS · NON-CALCULATOR

1 Row 1: 0.75 and 75%. Row 2: 7/20 and 35%. Row 3: 3/5 and 0.6. [6]

One mark per correct box. 35/100 not simplified to 7/20 loses that one box only.

2 (a) 45/100 = 9/20 [2]
(b) 24/100 = 6/25

3 (a) 7 ÷ 8 = 0.875 [2]
(b) 5 ÷ 16 = 0.3125

4 (a) 12/20 + 5/20 = 17/20 [4]
(b) 5/6 − 2/6 = 3/6 = 1/2
(c) 18/30 = 3/5
(d) 3/4 × 5/2 = 15/8 = 1 7/8

Award the mark for a correct unsimplified answer only if the question did not ask for simplest form — here it did.

5 (a) 120 [3]
(b) 9
(c) 30

(c) is 1/8 of 240. Halving three times is quicker than any other route.

6 Converting all four to decimals: 0.7, 0.6, 0.65, 0.667 [1] [2]
Order: 3/5, 65%, 2/3, 0.7 [1]

The second mark requires the original forms. Answering 0.6, 0.65, 0.667, 0.7 loses it.

7 1200 × 0.8 [1] [2]
= 960 rupees [1]

Finding 240 and stopping scores 1 — that is the discount, not the sale price.

8 40% of 30 = 12 wear glasses [1] [2]
30 − 12 = 18 [1]

Or 60% of 30 directly. Answering 12 scores 1: the question asked who does not.

9 (a) 500 ÷ 4 = 125, so 3 × 125 = 375 g [2] [3]
(b) 500 − 375 = 125 g [1]

10 (a) 18/24 = 3/4 [2]
(b) 75%

11 8000 × 1.05 [1] [2]
= 8400 rupees [1]

Paper B — Medium

30 MARKS · NON-CALCULATOR

1 (a) 13/5 [2]
(b) 4 1/4

2 (a) 3/2 + 11/4 = 6/4 + 11/4 = 17/4 = 4 1/4 [4]
(b) 10/3 − 11/6 = 20/6 − 11/6 = 9/6 = 1 1/2
(c) 5/4 × 12/5 = 60/20 = 3 [2]

In (c), one mark for converting both mixed numbers to improper fractions, one for the answer.

3 Decimals: 0.46, 0.444, 0.43, 0.45 [1] [2]
Order: 43%, 4/9, 9/20, 0.46 [1]

4 250 × 1.12 [1] [2]
= 280 [1]

5 640 × 0.85 [1] [2]
= 544 [1]

6 Change = 800 − 680 = 120 [1] [3]
120 ÷ 800 [1]
= 15% decrease [1]

Dividing by 680 instead of 800 gives 17.6% — the single most common error in this topic. Award the first mark only.

7 Change = 4600 − 4000 = 600 [1] [3]
600 ÷ 4000 [1]
= 15% increase [1]

8 320 ÷ 8 = 40, so 3 × 40 [1] [2]
= 120 [1]

9 5/8 of 400 = 250 [1] [3]
60% of 420 = 252 [1]
60% of 420 is larger (by 2) [1]

The third mark is for the conclusion in words. Two correct numbers with no statement scores 2.

10 68% of 250 = 170 full [1] [2]
250 − 170 = 80 empty [1]

Or 32% of 250 directly.

11 375/1000 = 3/8 [1] [2]
37.5% [1]

12 Boys = 40 − 24 = 16 [1] [3]
16/40 [1]
= 40% [1]

Answering 60% means the student found the girls. That scores nothing beyond the method mark.

Paper C — Hard

30 MARKS · NON-CALCULATOR

1 (a) Terminates, 0.1875 (16 = 2×2×2×2) [3]
(b) Recurs, 0.41666… (12 has a factor of 3)
(c) Terminates, 0.28 (25 = 5×5)

Both halves needed for each mark. A fraction terminates only when the denominator, in simplest form, has no prime factors other than 2 and 5.

2 The multiplier for a 20% increase is 1.2 [1] [3]
Original = 96 ÷ 1.2 [1]
= 80 rupees [1]

Taking 20% off 96 gives 76.80 and scores zero. You must divide, not subtract.

3 The multiplier for a 25% decrease is 0.75 [1] [3]
Original = 180 ÷ 0.75 [1]
= 240 rupees [1]

4 (a) Starting from 100: 100 × 1.1 = 110, then 110 × 0.9 = 99, which is not 100 [2] [3]
(b) 1.1 × 0.9 = 0.99, so an overall 1% decrease [1]

Any starting value is acceptable in (a). The second 10% is taken off a larger number than the first was added to, so the two never cancel.

5 3/5 of 450 = 270 [1] [3]
2/3 of 270 [1]
= 180 [1]

Also acceptable: 2/3 × 3/5 = 2/5, then 2/5 of 450 = 180. Full marks for either route.

6 (a) Shop A: 2400 × 0.7 = 1680 [1] [4]
Shop B: 2400 × 0.75 = 1800, then 1800 × 0.9 = 1620 [1]
(b) Shop B is cheaper [1], by 1680 − 1620 = 60 rupees [1]

Adding 25% and 10% to get 35% off is the trap. Successive percentage changes multiply.

7 5/8 of 640 = 400 [1] [3]
15% of 400 [1]
= 60 [1]

8 Losing 12% leaves 88%, so the multiplier is 0.88 [1] [3]
4.4 ÷ 0.88 [1]
= 5 kg [1]

Adding 12% to 4.4 gives 4.93 and scores at most the first mark.

9 Recognising a single recurring digit gives ninths [1] [2]
= 4/9 [1]

Worth memorising: 0.111… = 1/9, so 0.444… = 4/9 and 0.777… = 7/9.

10 Overall multiplier = 0.65 × 0.9 = 0.585 [1] [3]
So the customer pays 58.5% of the original, a discount of 41.5% [1]
The sign is wrong because the two discounts have been added; the second 10% is taken off an already reduced price, so it is worth less than 10% of the original [1]

Paper D — Hard

30 MARKS · NON-CALCULATOR

1 Profit = 600 − 480 = 120 [1] [3]
120 ÷ 480 [1]
= 25% profit [1]

Percentage profit is always measured against the cost price, never the selling price.

2 A 20% loss means he received 80%, so the multiplier is 0.8 [1] [3]
640 ÷ 0.8 [1]
= 800 rupees [1]

3 (a) After books he has 2/3 left. He then spends 1/4 of that, so he keeps 3/4 of 2/3 [1] [4]
3/4 × 2/3 = 1/2 [1]
(b) 1/2 of the original = 900 [1], so he started with 1800 rupees [1]

The trap is adding 1/3 and 1/4 to get 7/12 spent. The second fraction is of the remainder, not of the original.

4 Fraction removed = 3/5 − 1/3 [1] [4]
= 9/15 − 5/15 = 4/15 [1]
4/15 of the capacity = 240, so 1/15 = 60 [1]
Capacity = 60 × 15 = 900 litres [1]

5 Deal 1: 600 × 0.8 × 0.9 = 432 [2] [4]
Deal 2: 600 × 0.75 × 0.95 = 427.50 [1]
Deal 2 is better, by 4.50 rupees [1]

Both deals total 30% if you add them, which is exactly why they look identical and are not.

6 Take the original as 100; after a 25% rise it is 125 [1] [4]
The fall needed is 25 out of 125 [1]
25 ÷ 125 = 0.2 [1]
So the price must fall by 20% [1]

Answering 25% scores at most one mark. The fall is measured against the new, larger price — this is the same idea as the 10% up-and-down question on Paper C.

7 Girls = 3/5 of 500 = 300, so boys = 200 [1] [4]
25% of 300 = 75 girls walk [1]
40% of 200 = 80 boys walk [1]
Total = 75 + 80 = 155 students [1]

Averaging 25% and 40% to get 32.5% of 500 gives 162.5 and scores nothing. The two groups are different sizes.

8 (i) The mistake: comparing decimals by the size of the digits after the point instead of by place value [1] [4]
Padding with a zero gives 0.90 against 0.15, so 0.9 is larger [1]
(ii) The mistake: adding the numerators and the denominators separately [1]
Using a common denominator: 3/6 + 2/6 = 5/6 [1]

A useful check for (ii): the answer must be bigger than 1/2, and 2/5 is smaller — so it is wrong before you calculate anything.